

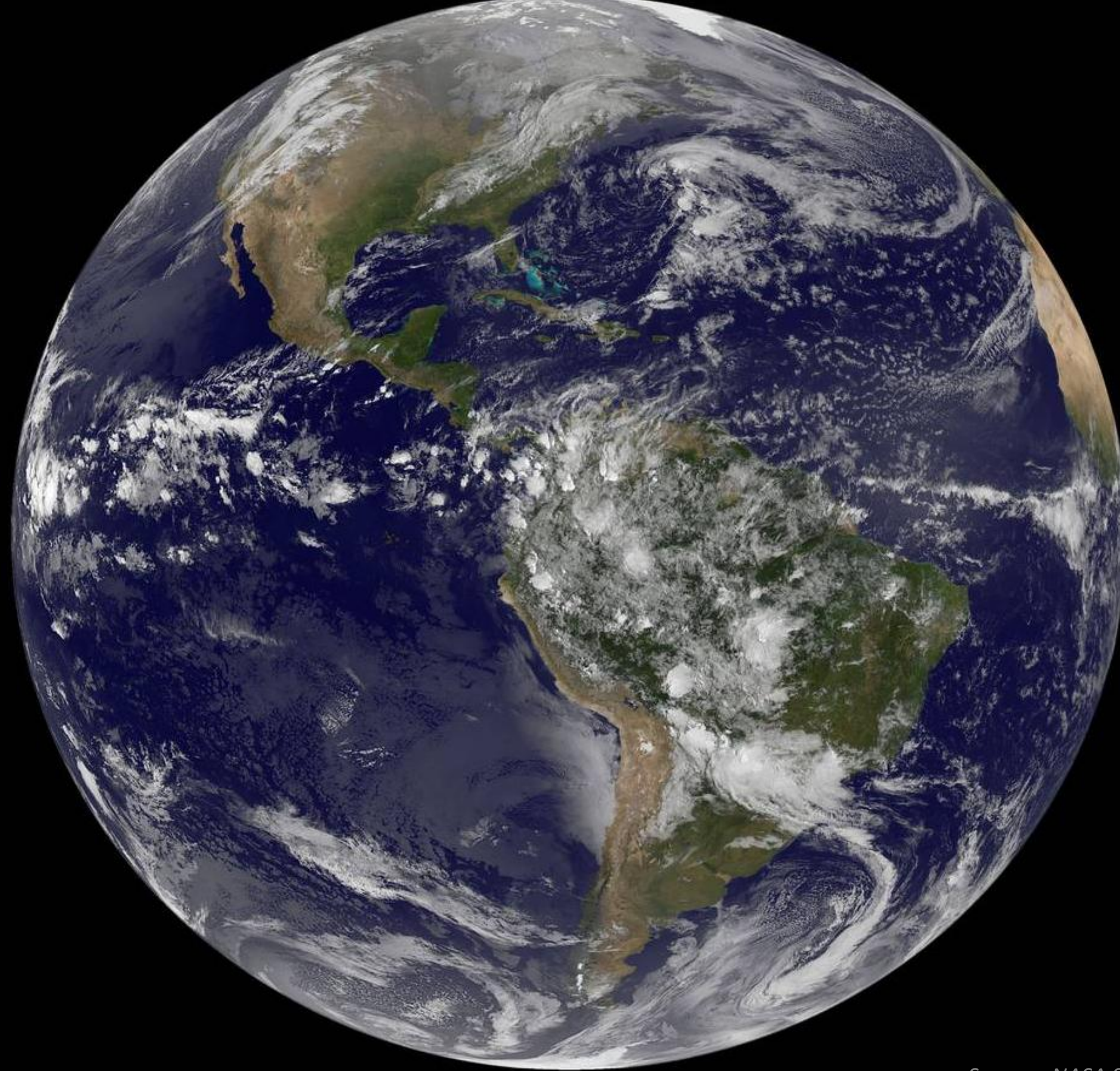
Deep learning-based Cloud-robust Waterbody detection

Hydro AI

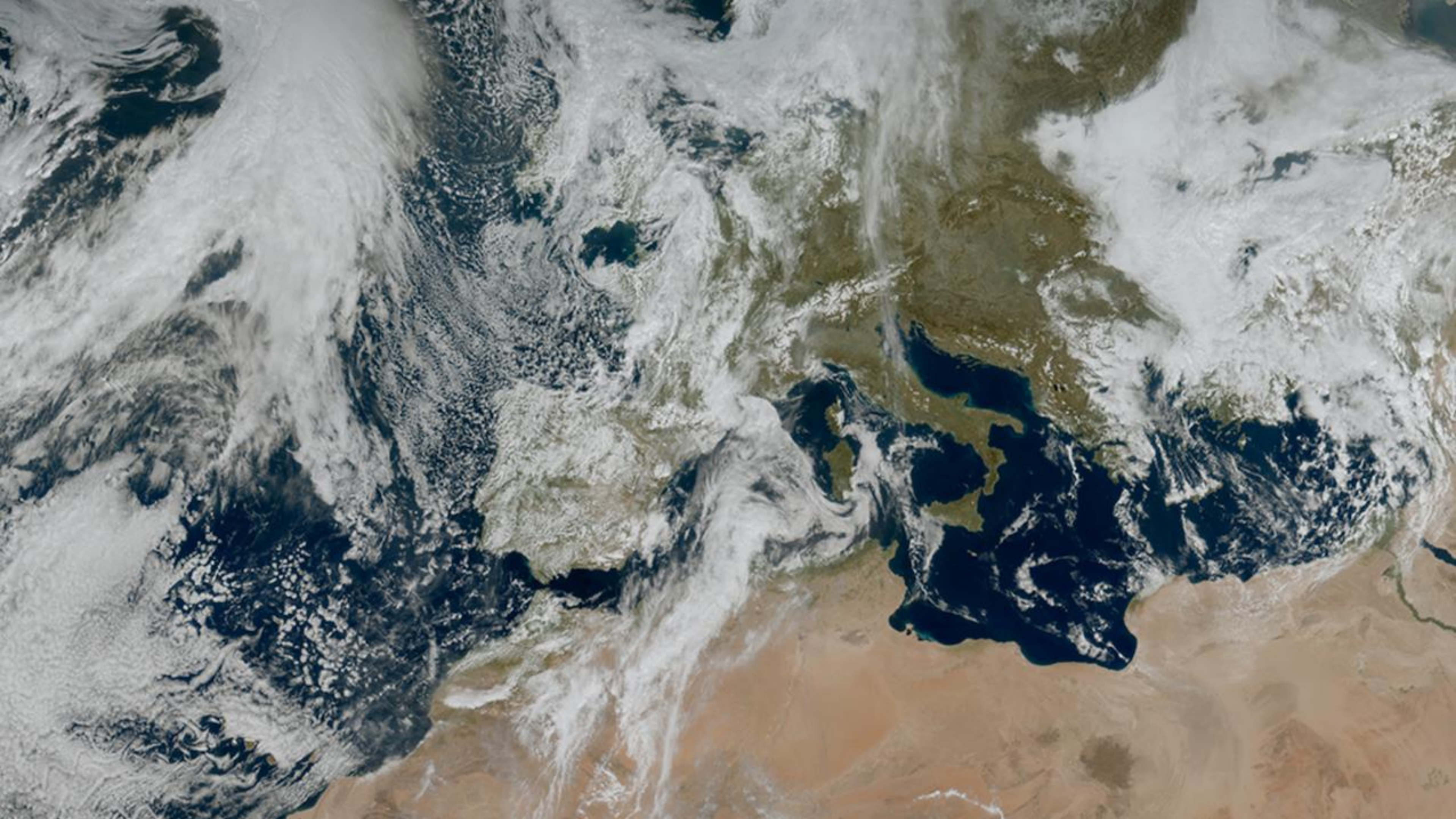
Beomsik Kim

Prof. Hyunglok kim





Source : NASA Satellite View of the Americas on Earth Day



0. Introduction



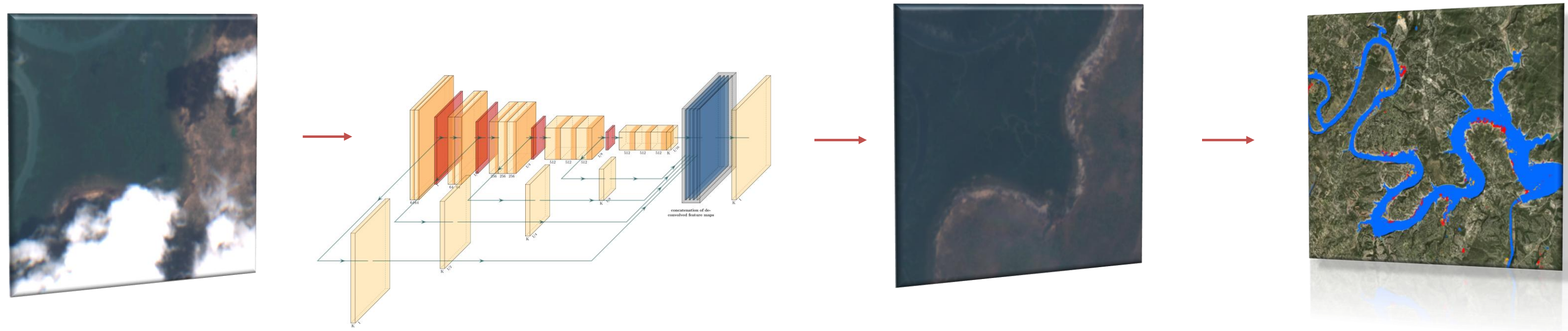
- AI can forecast hydrological processes, including snowpack dynamics, soil moisture and river flows

Eyring et al., Nat. Geosci. (2024)

- Water bodies extraction is critical for hydrology and water resources management

J. Indian Soc. Remote Sens. (2023)

0. Introduction



- AI can forecast hydrological processes, including snowpack dynamics, soil moisture and river flows

Eyring et al., Nat. Geosci. (2024)

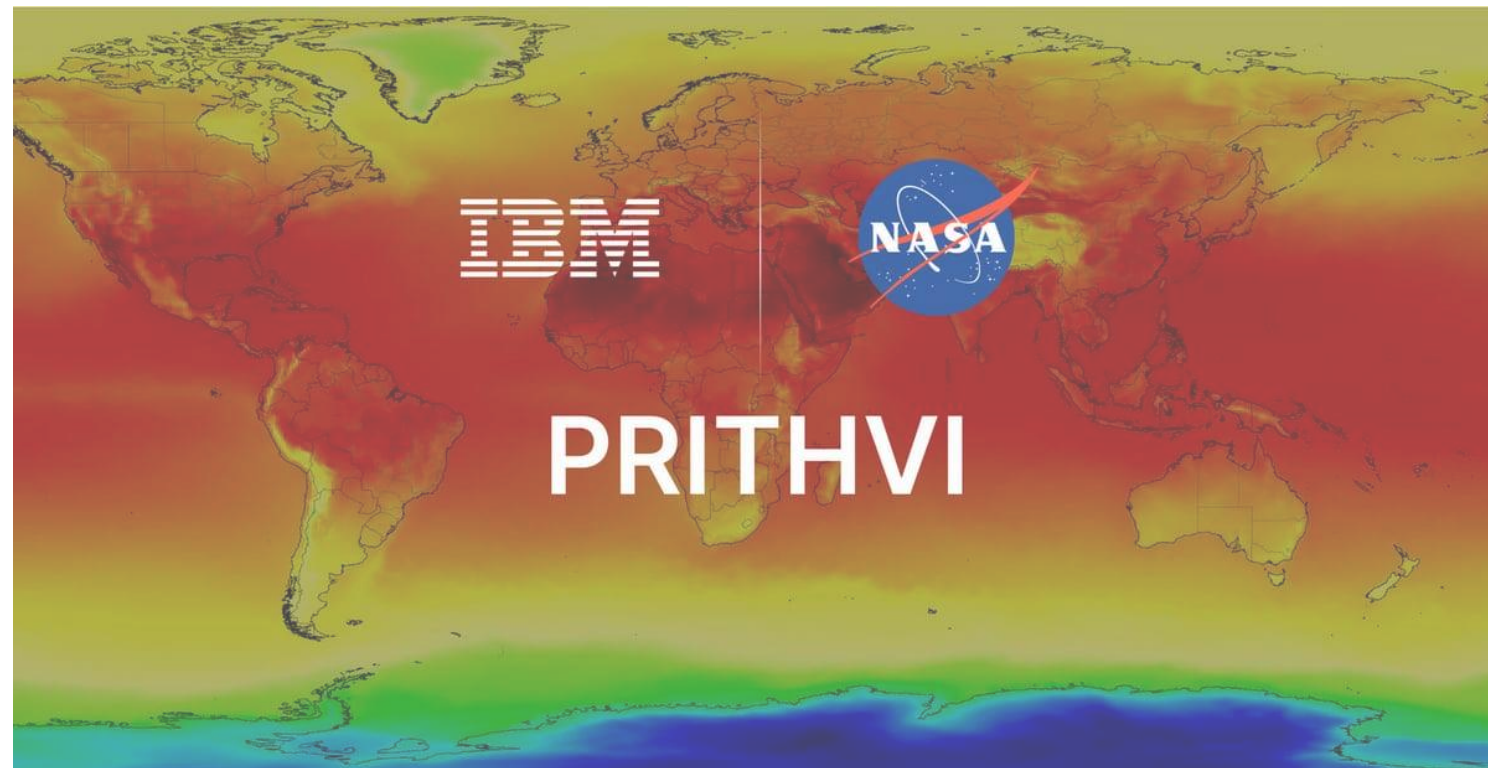
- Cloud contamination significantly impairs the usability of optical satellite imagery, affecting critical applications such as **environmental monitoring**, disaster response, and landuse analysis.

Cloud-Aware SAR Fusion for Enhanced Optical Sensing in Space Missions ,Trong-An Bui(2025)

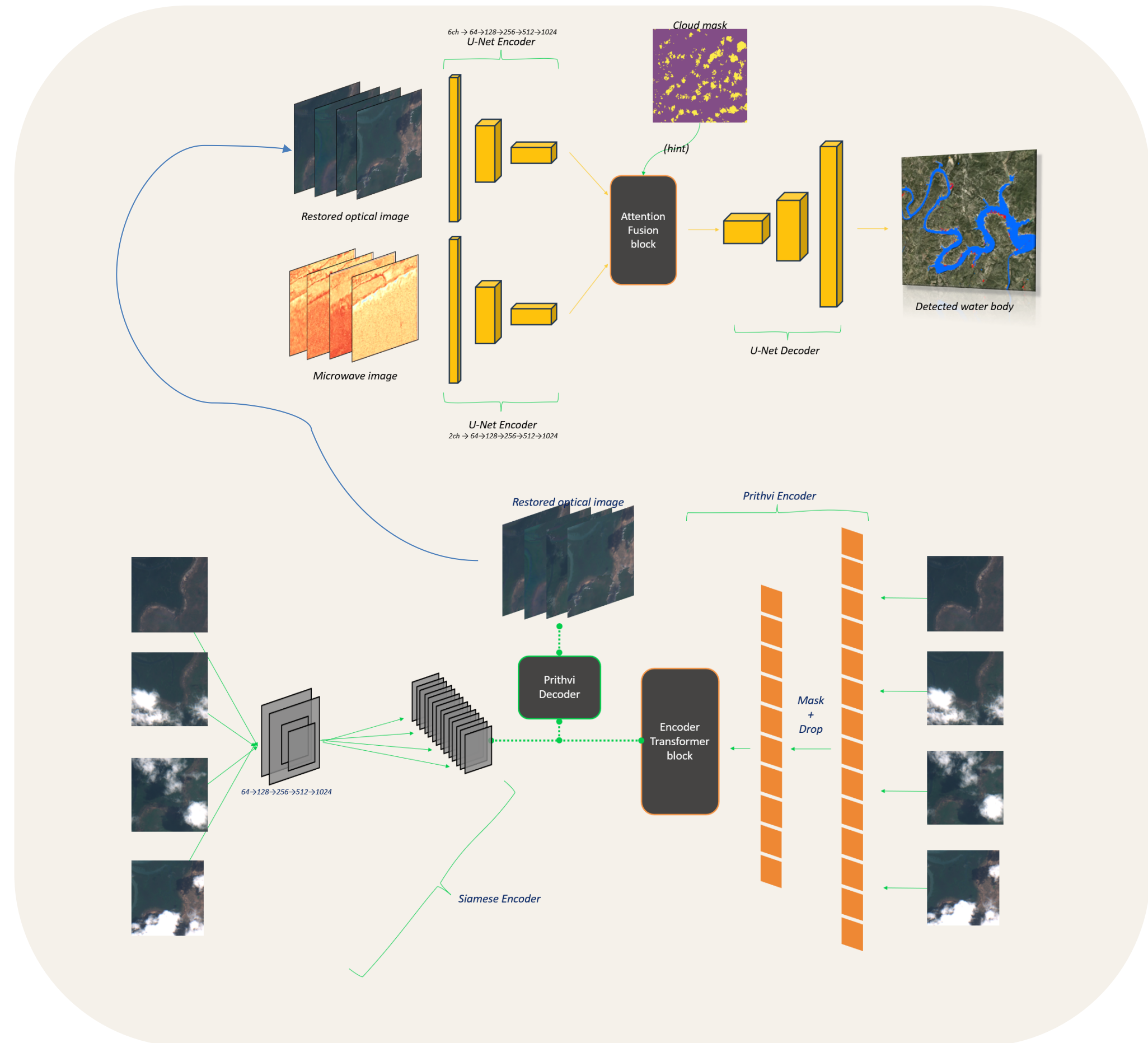
1. Research Goal

- 1 Optical Satellite Image Restoration for Water Body Detection.
- 2 Building a robust water body detection model for the 'amount' of clouds

2. Model Architecture

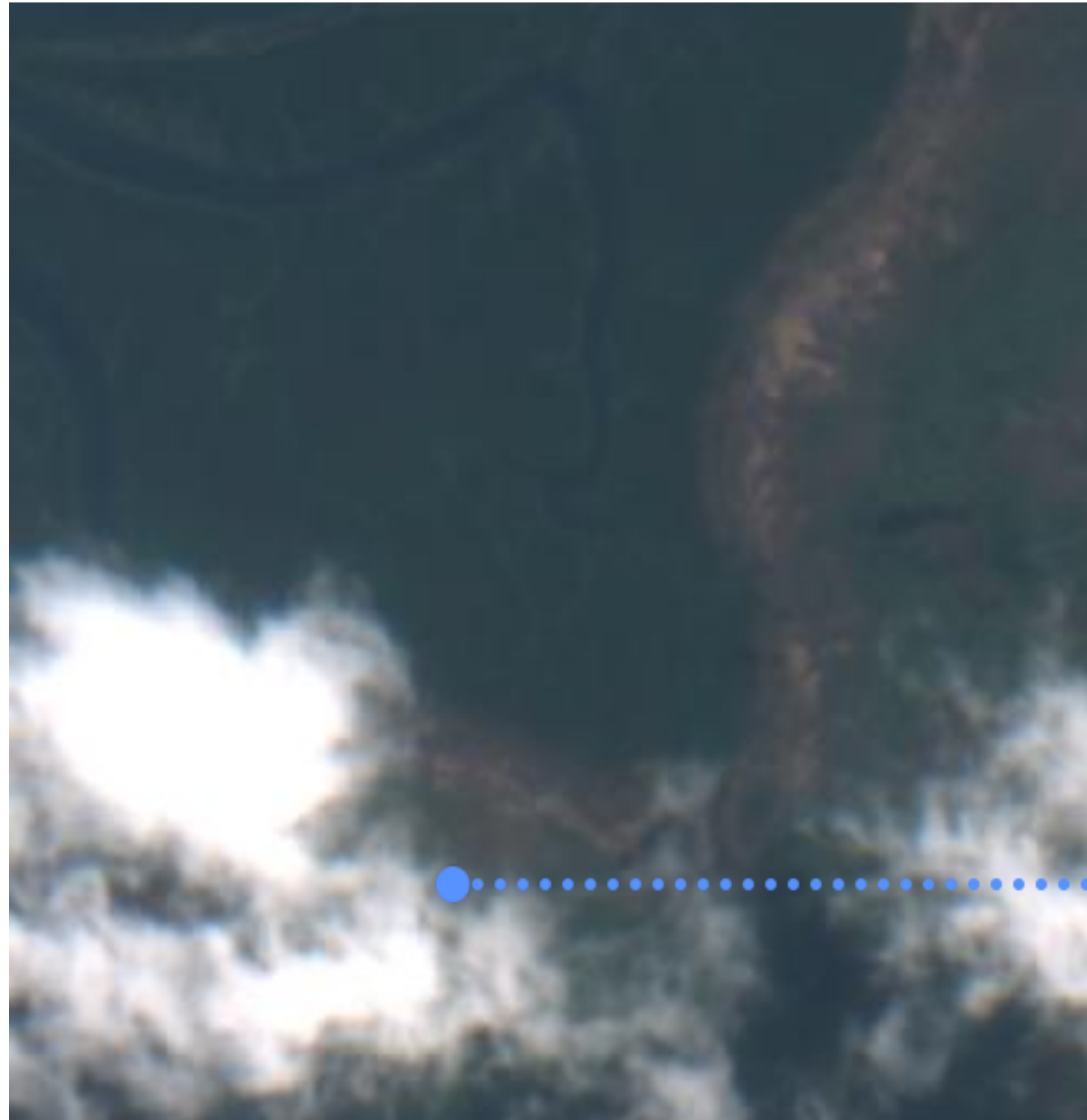


- **Prithvi** is temporally and spatially understood by the model through temporal and spatial attentions

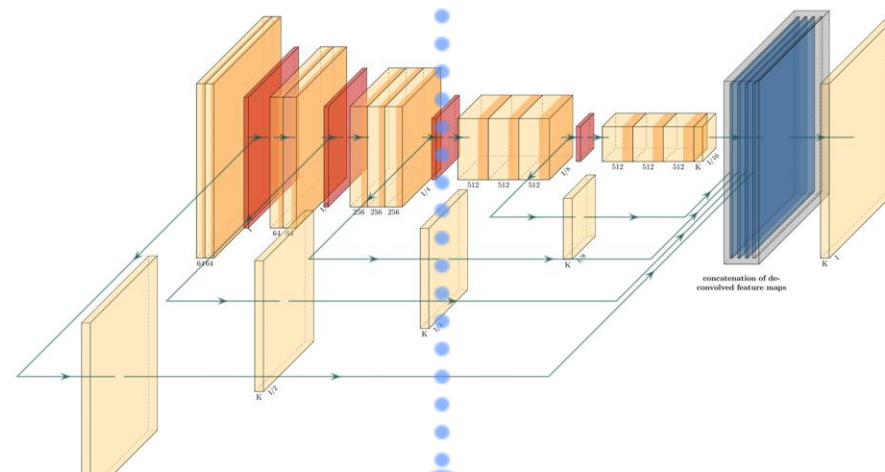


Results : Cloud Removal

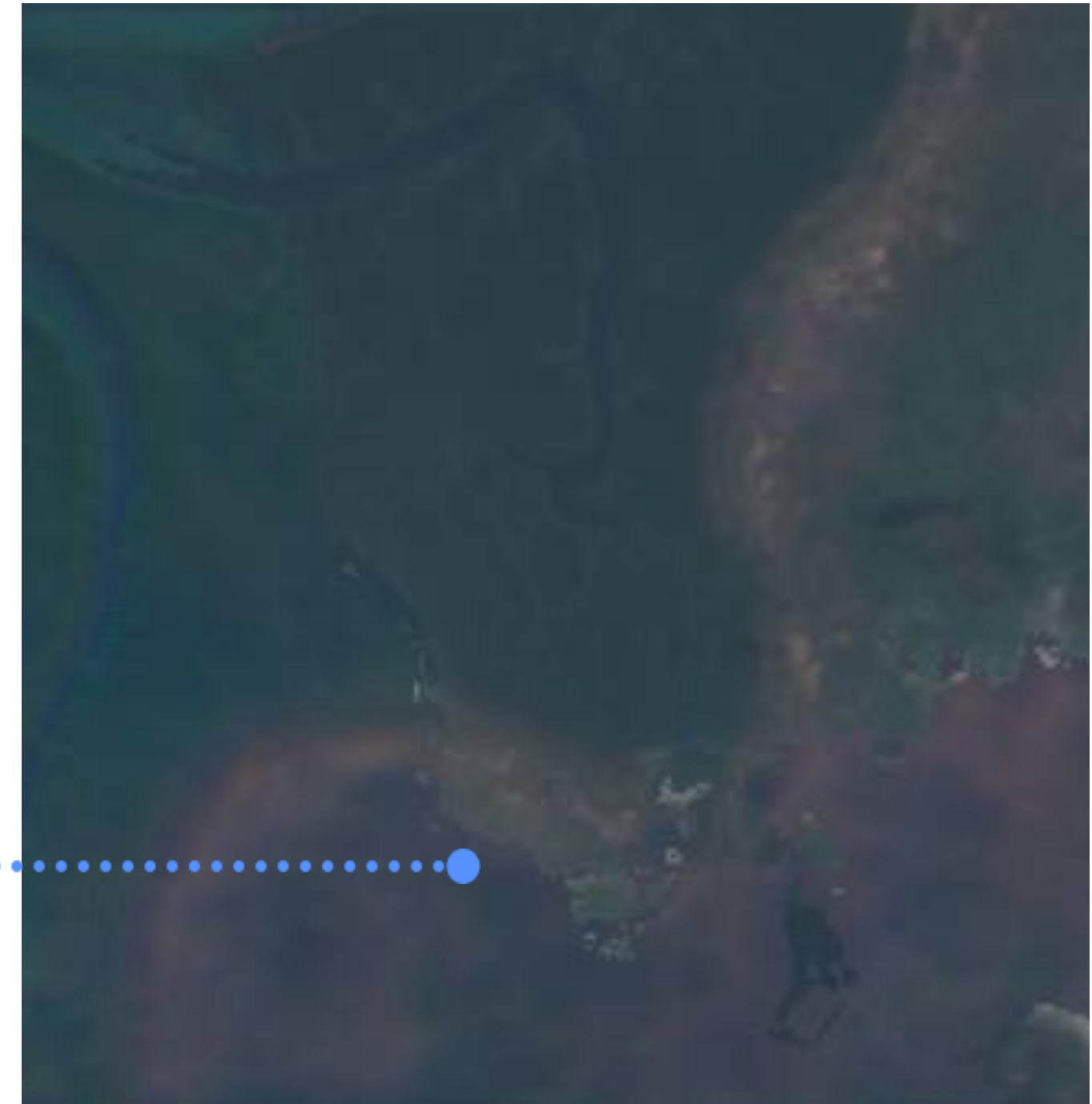
3. Result – restore cloud area



Before



A geographical boundary successfully restored.



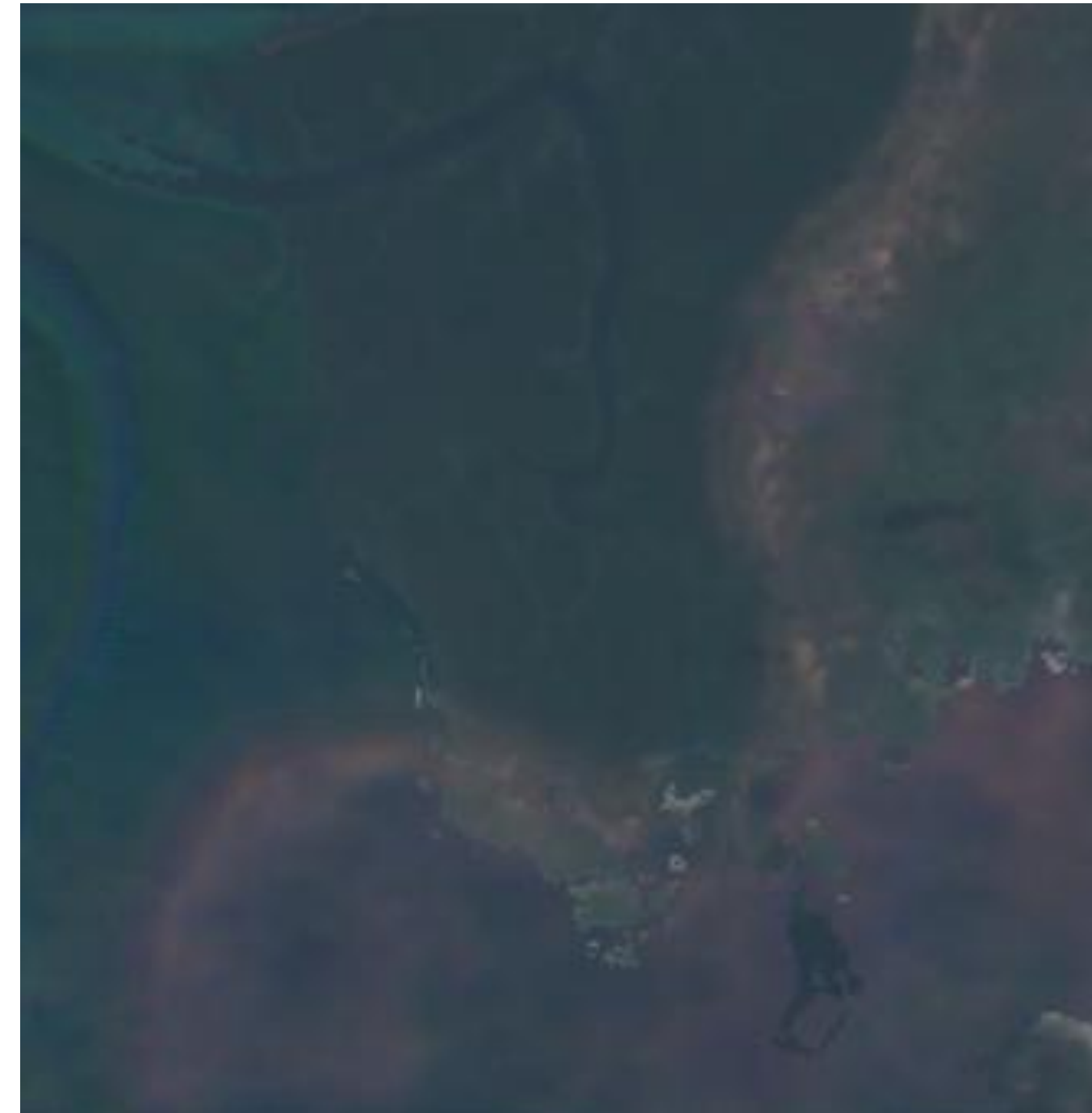
After

3. Result – restore cloud area



West Anhem, Australia

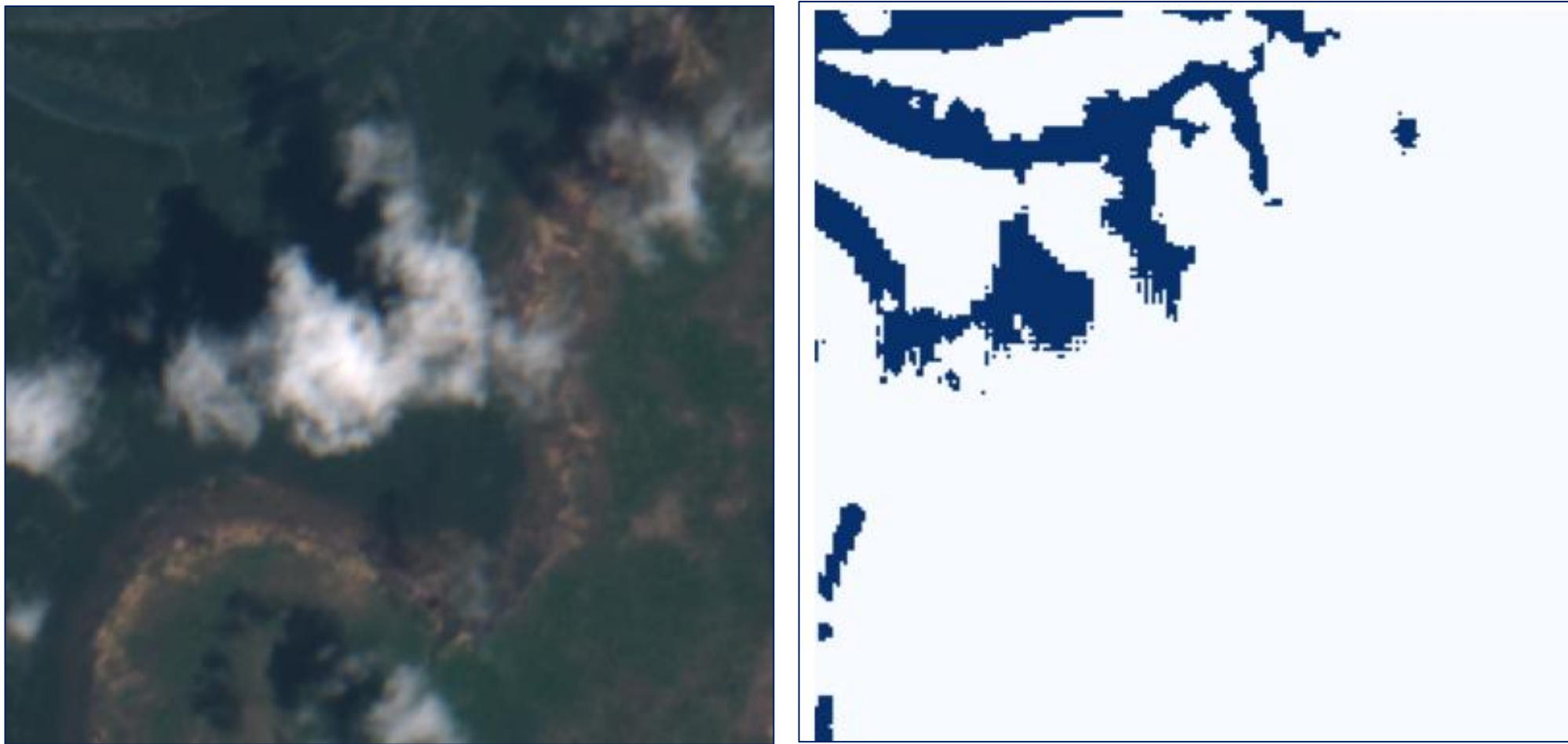
Real map



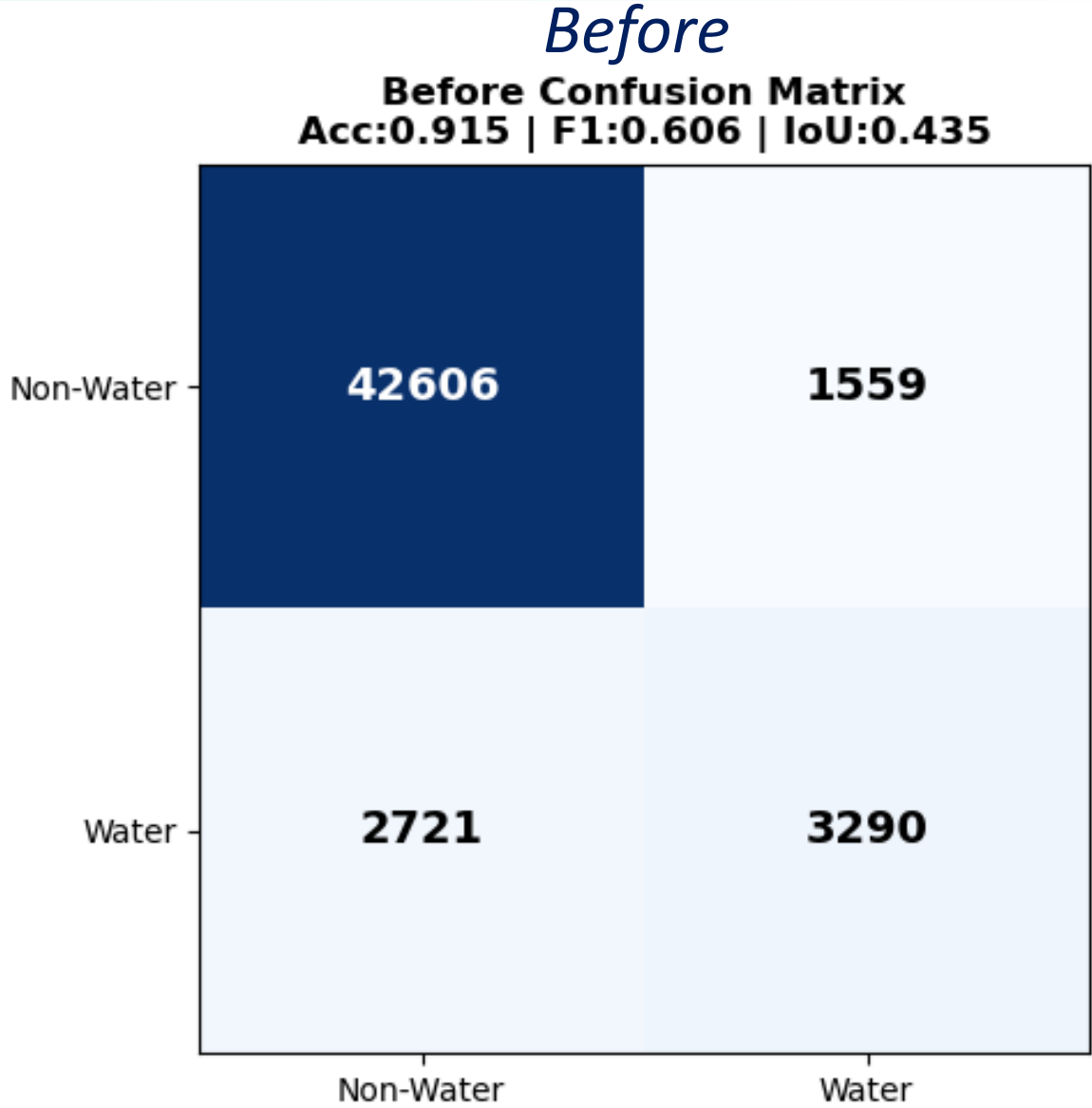
After

Results : Water-Body Detection

3. Result – restore cloud area / water body detection



Real map



$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

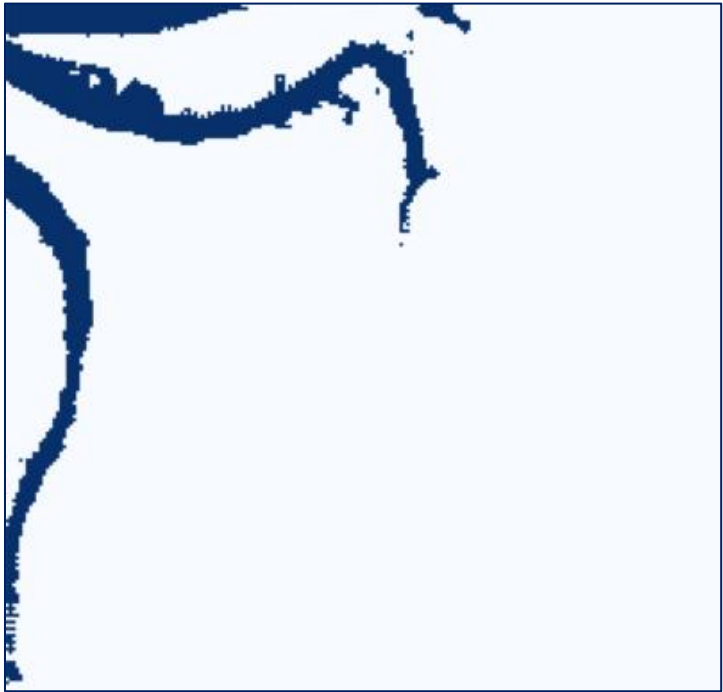
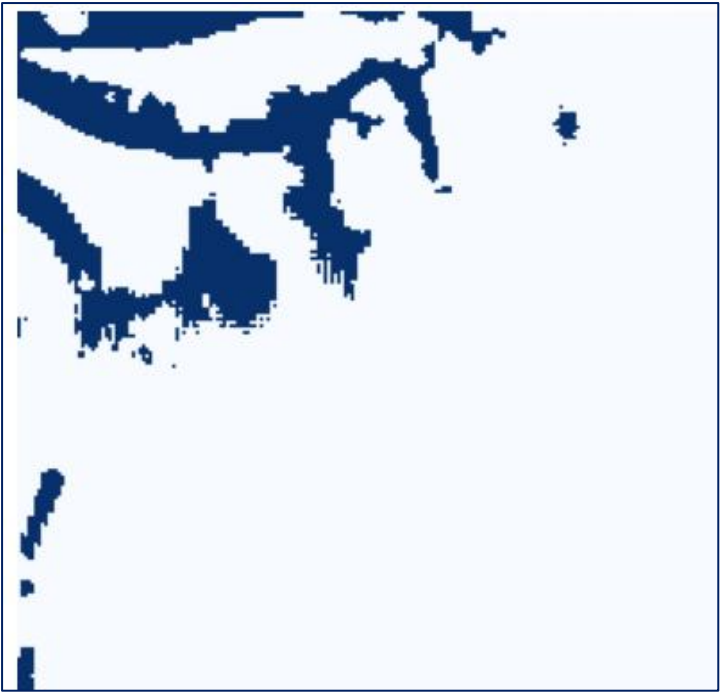
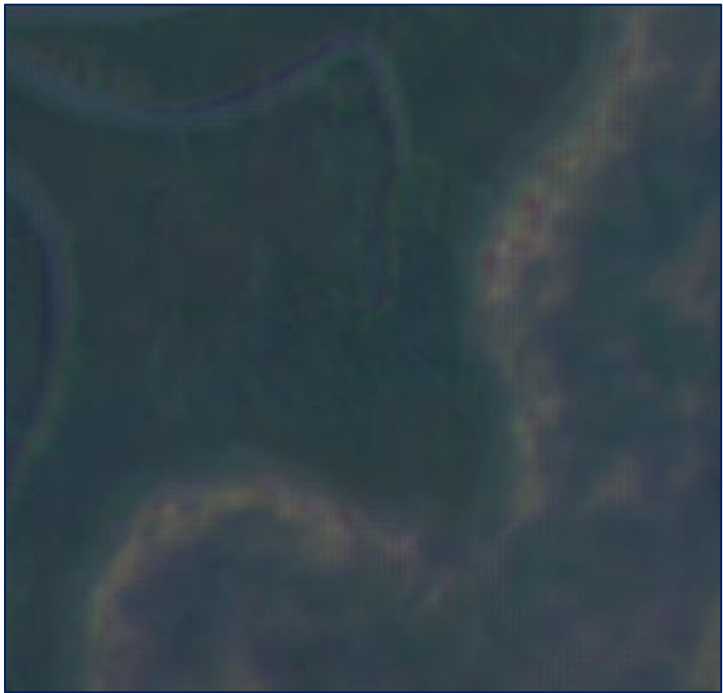
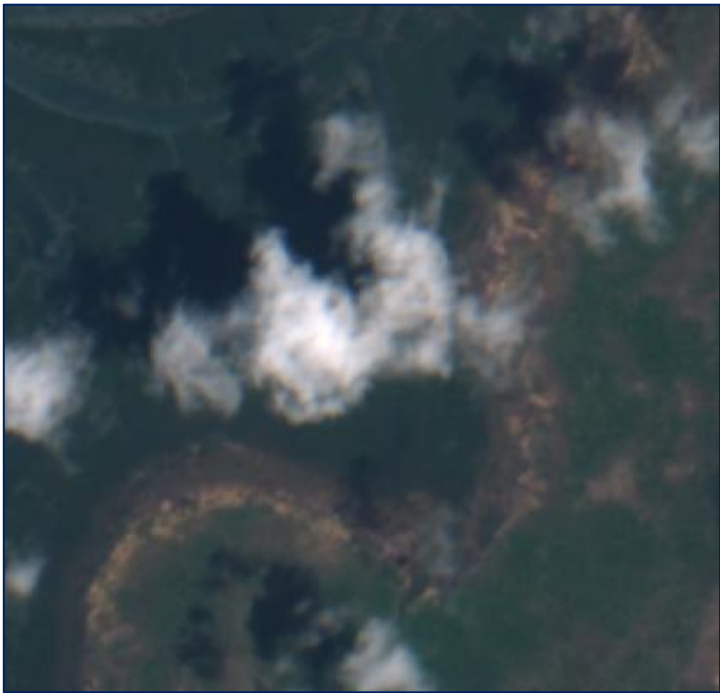
Real water

Model predict

3. Result – restore cloud area / water body detection

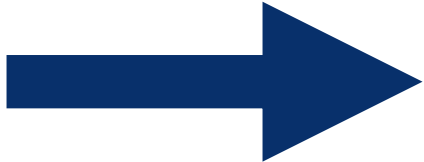
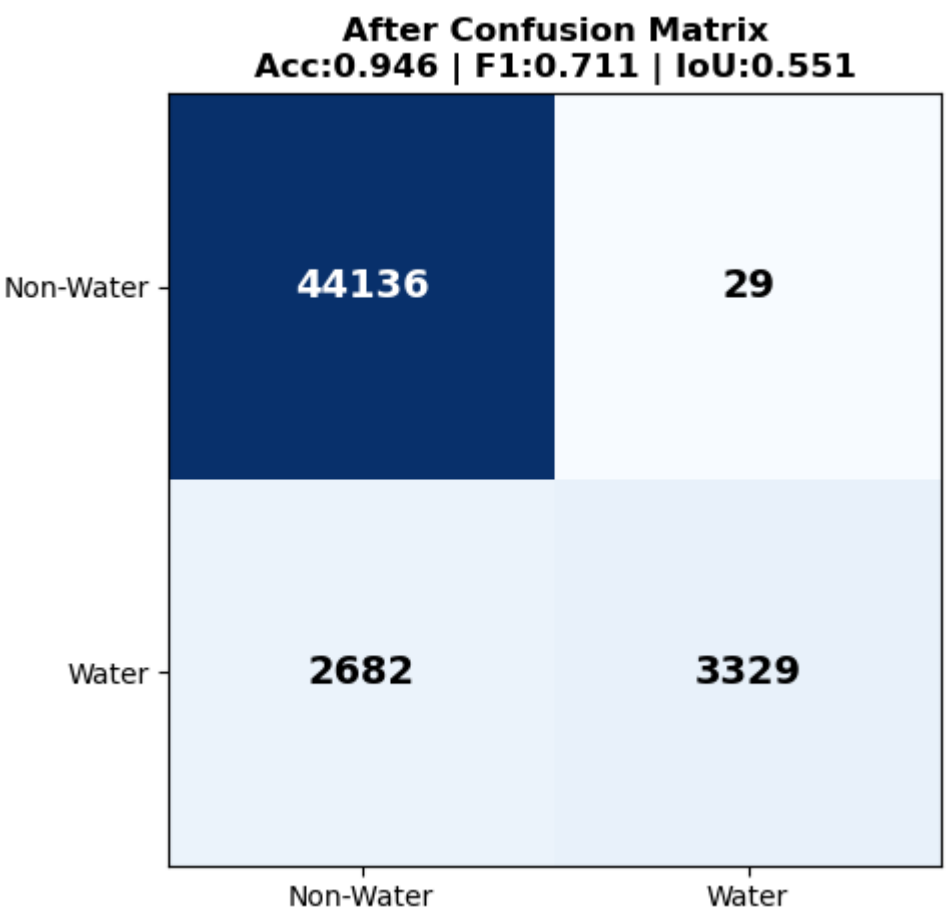
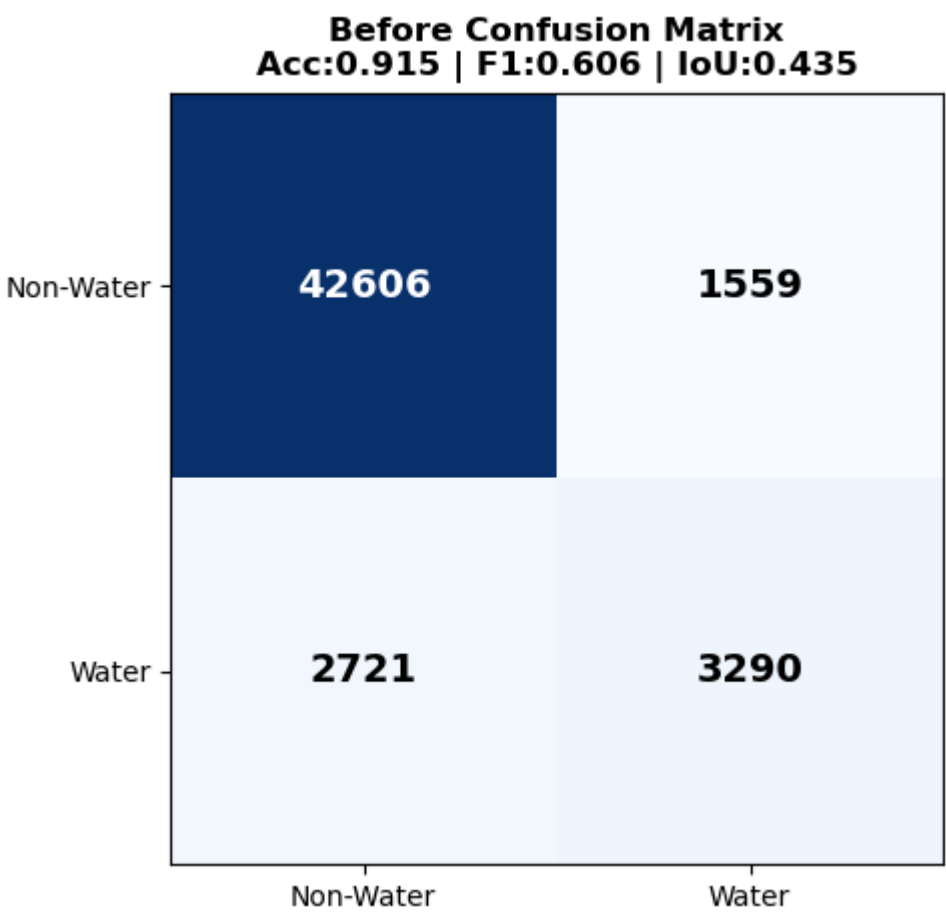
Before

After



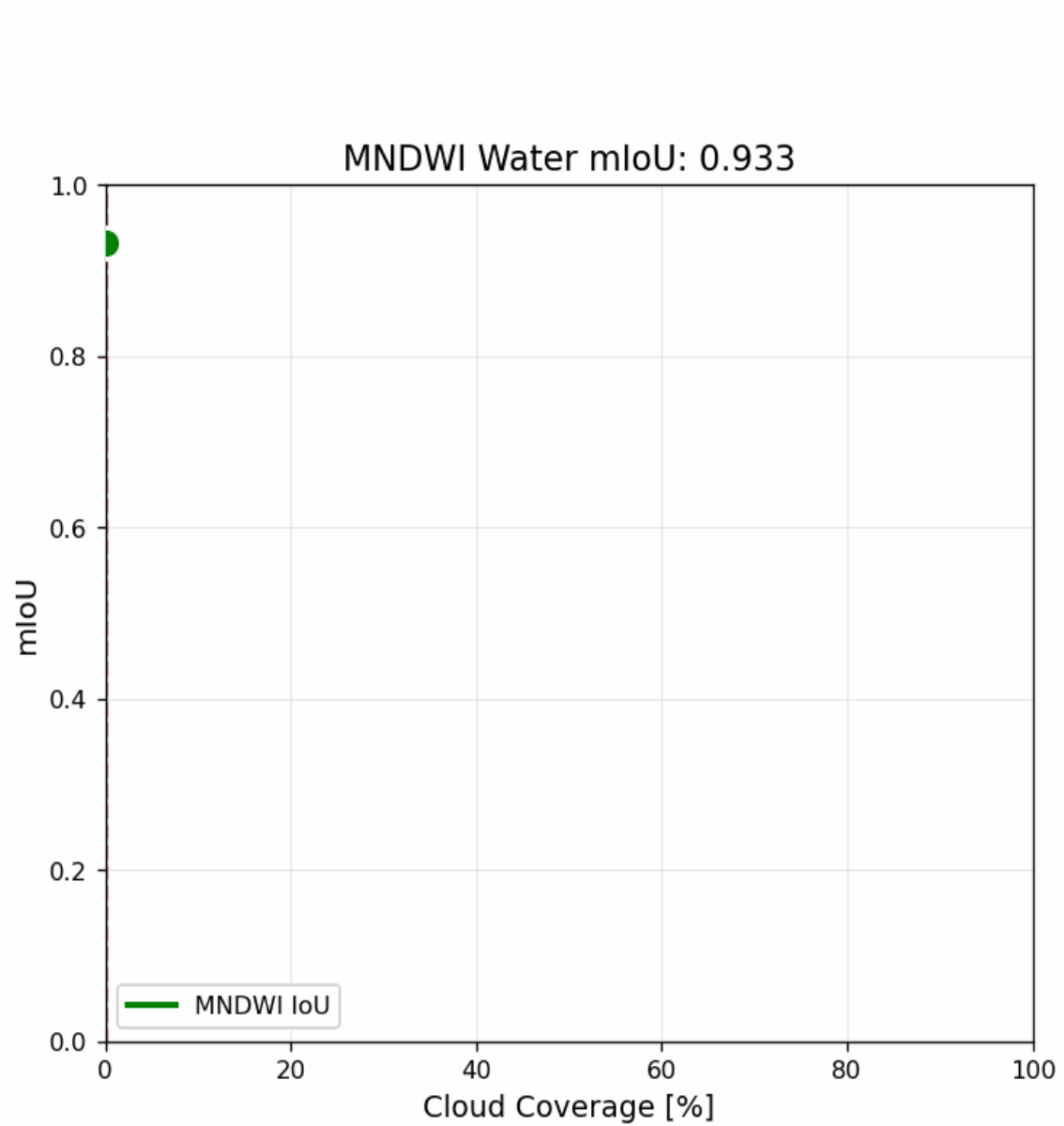
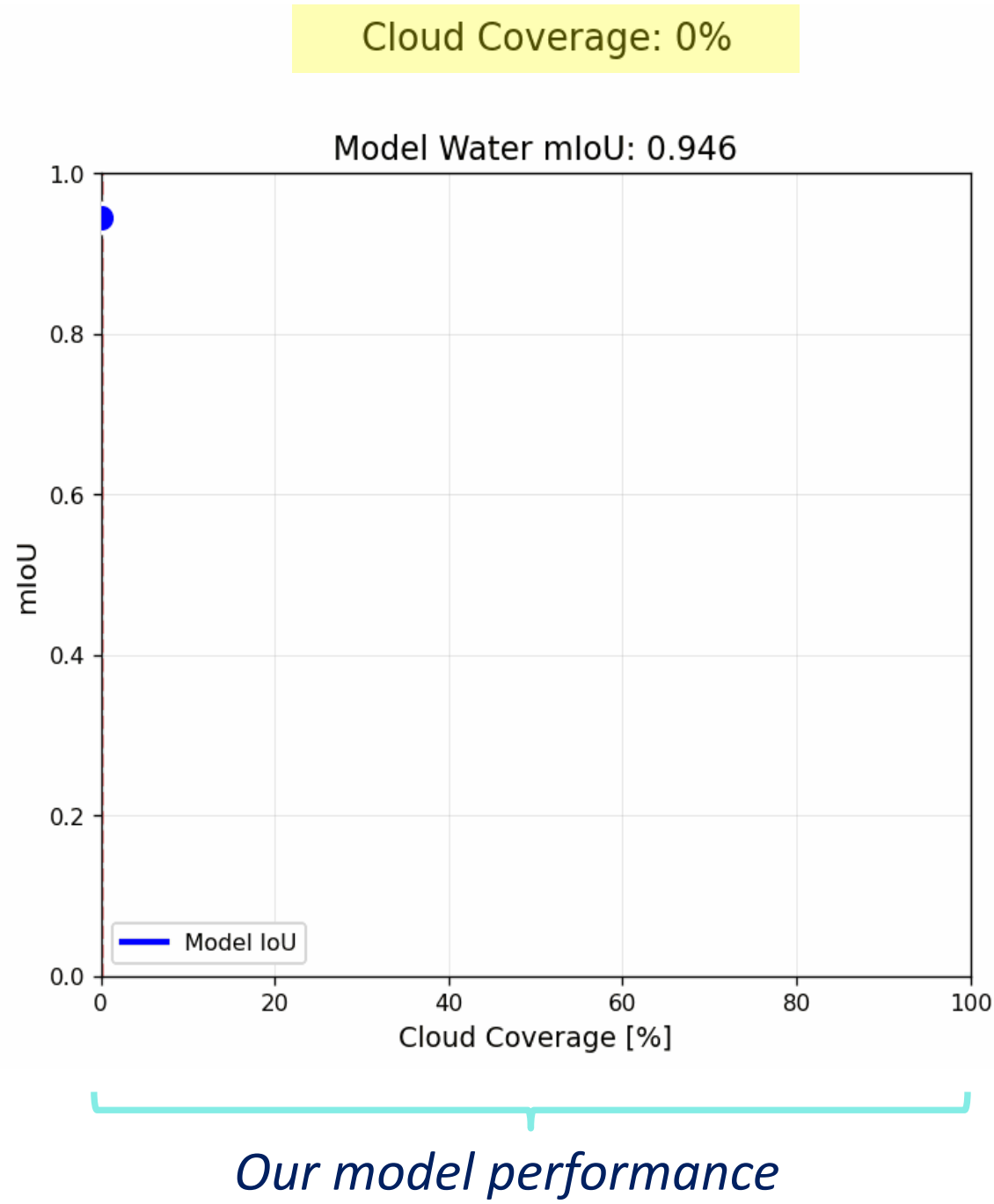
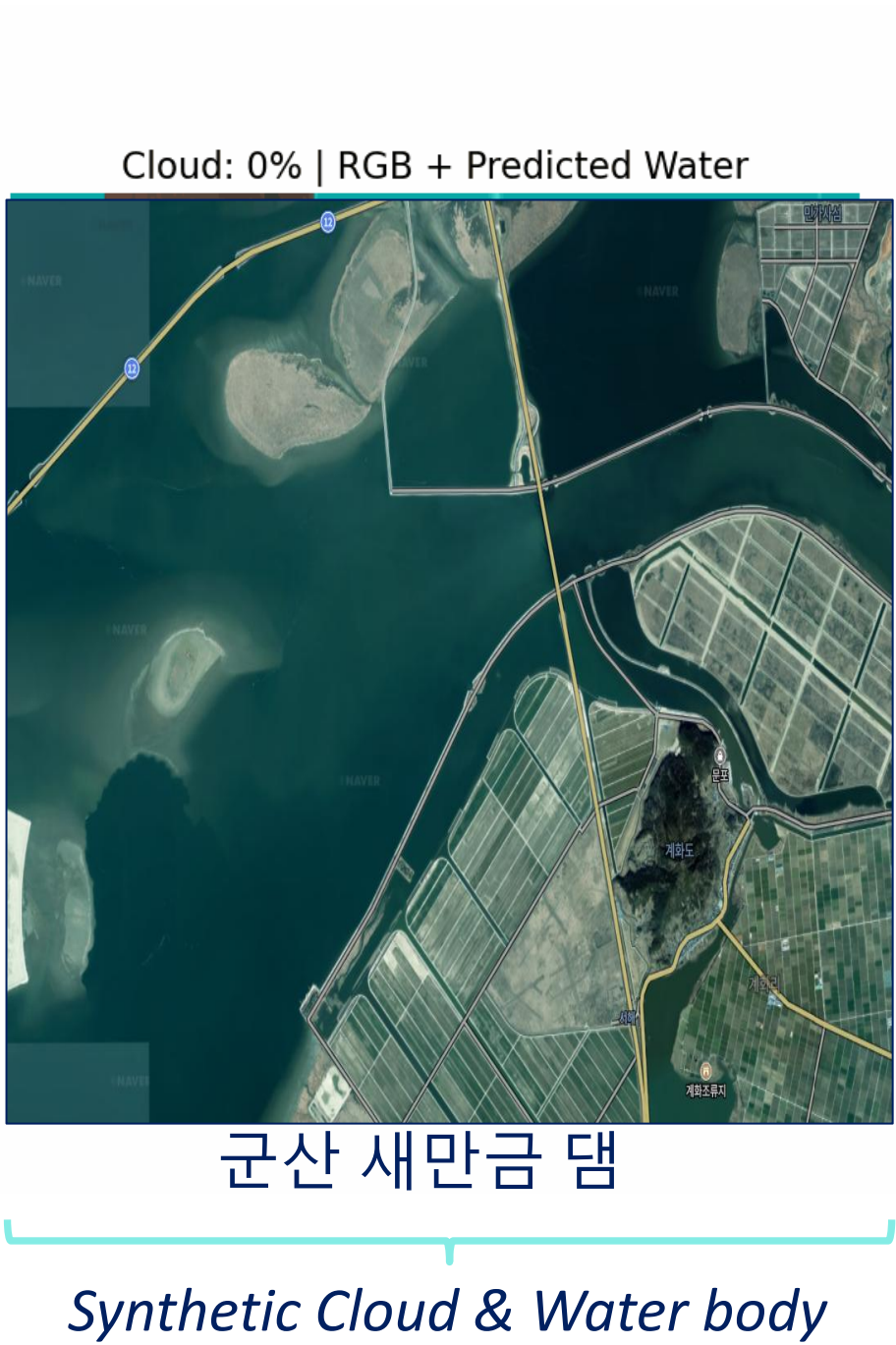
Before

After



Water IoU 26.7% improvement

3. Result – cloud amount experiment



4. Conclusion

- 수정필요

Strengths

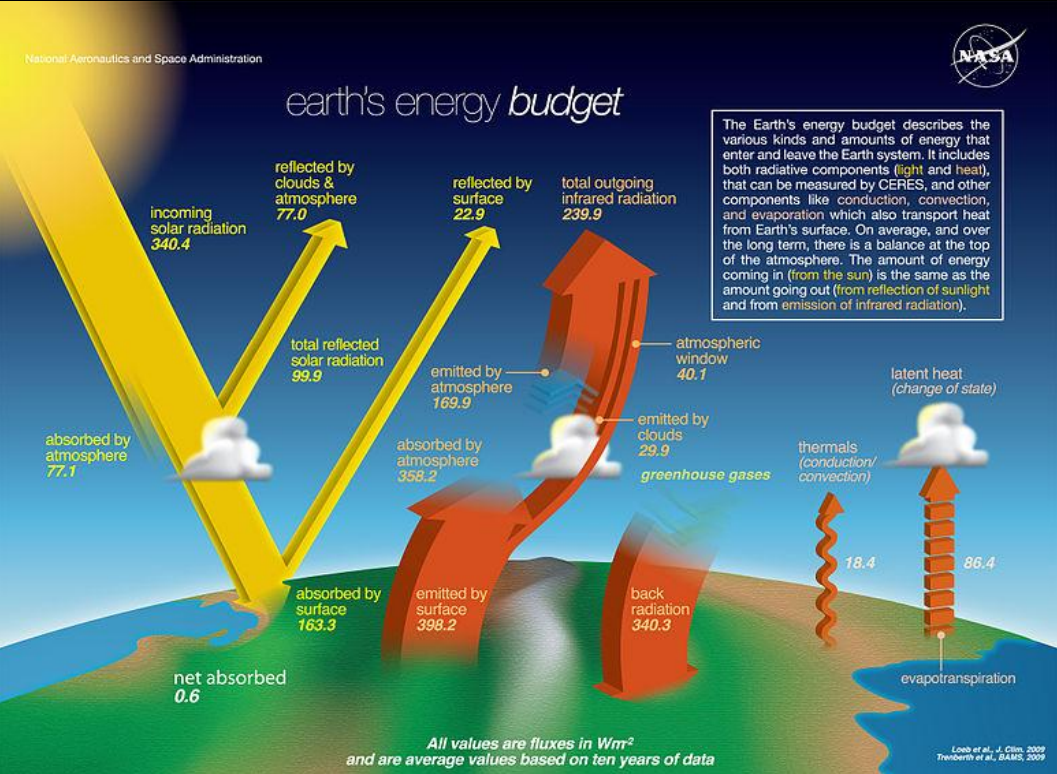
- Robust to cloud coverage, applicable to various downstream tasks
- Beyond simple image generation: leverages temporal context and terrain understanding for prediction

Limitation

- Output is model-generated, not actual cloud-free observation — may not be suitable for research requiring precise radiometric accuracy
- Temporal gap (1–2 days) between Sentinel-1 and Sentinel-2 acquisitions limits applicability for rapid flood dynamics or time-critical events

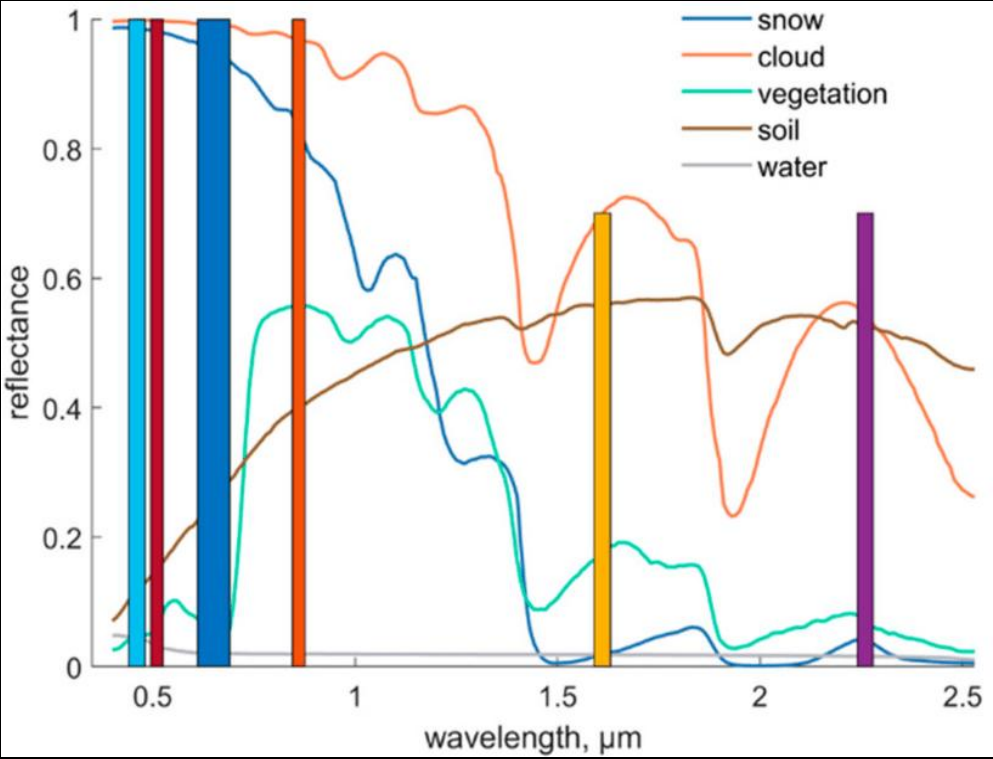
Thank you
Q&A



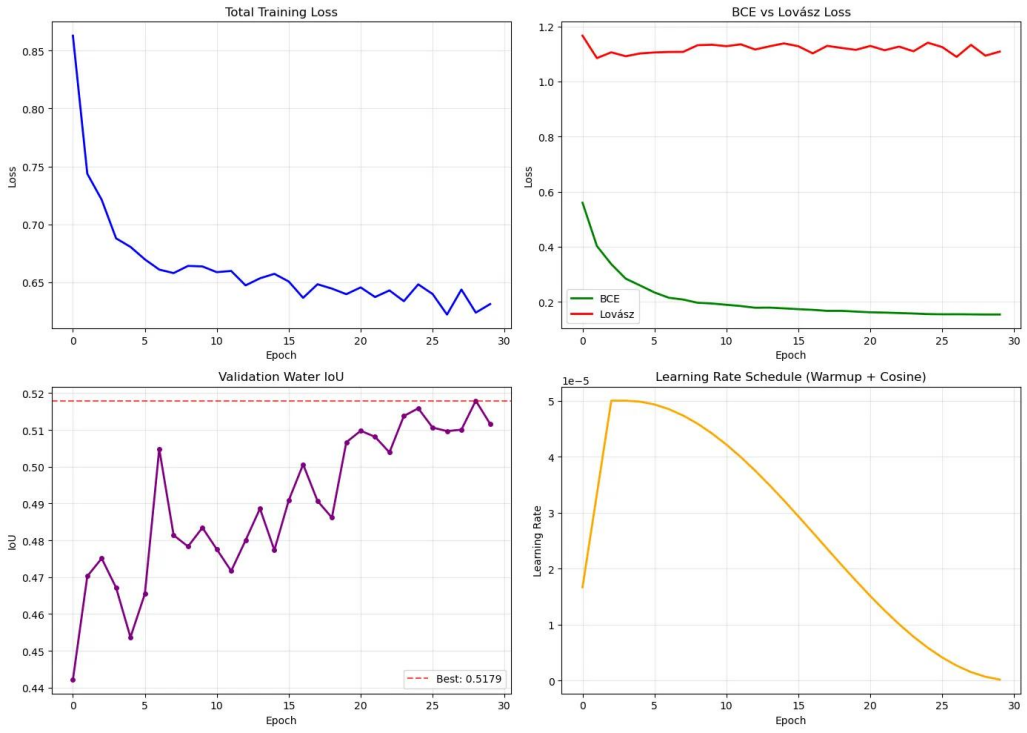
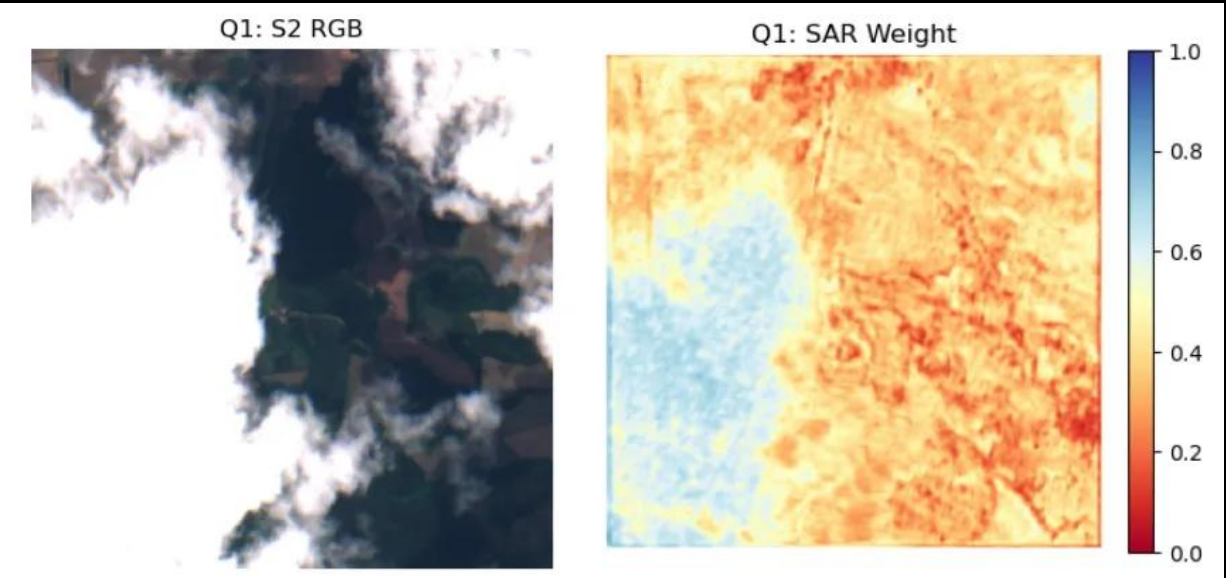


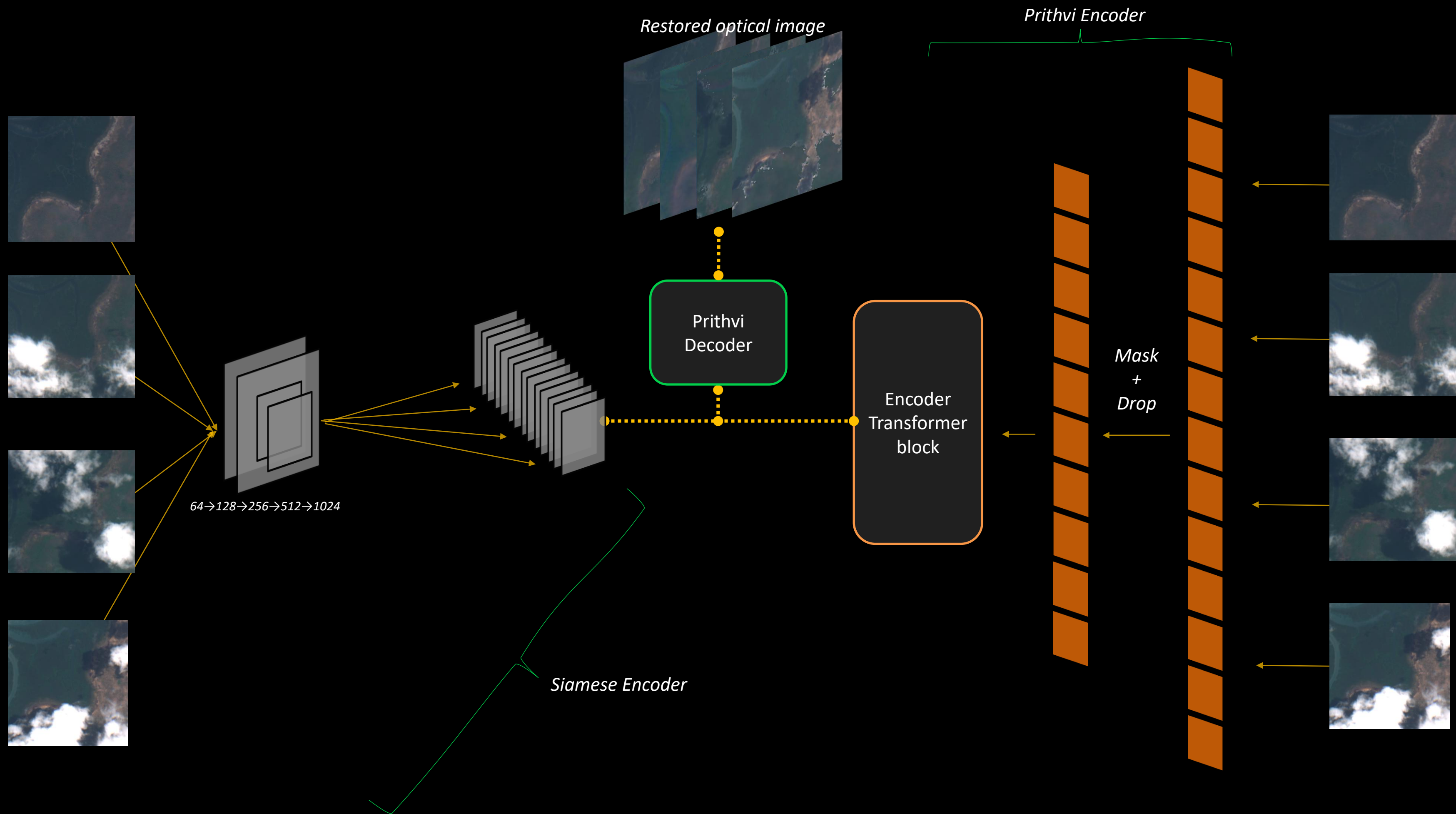
Earth's energy budget (Trenberth et al., 2009)

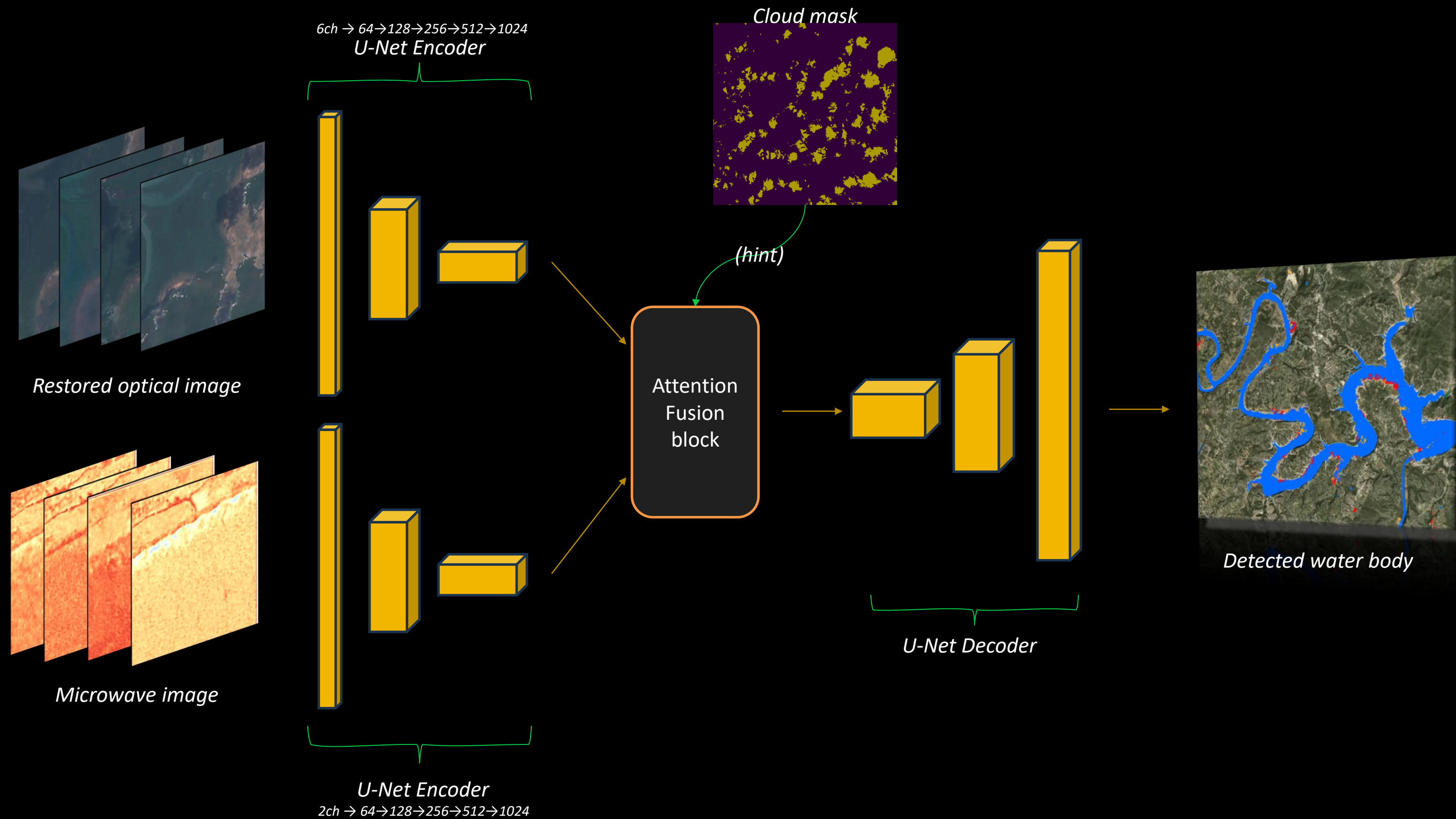
supplement

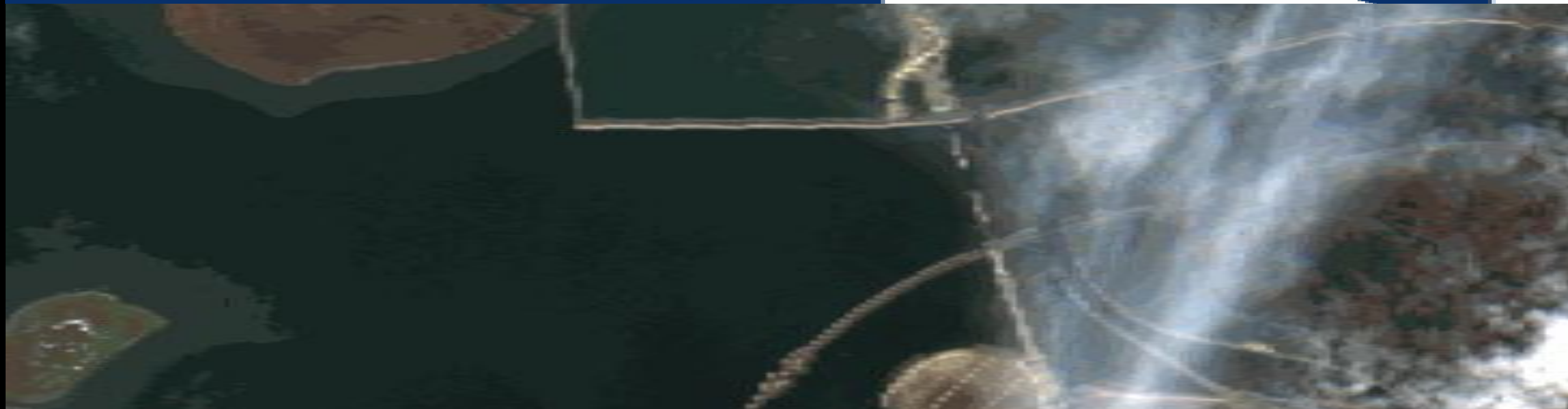


Reflectance spectral of water cloud, ice cloud, snow surface(Gao et al., 1998).

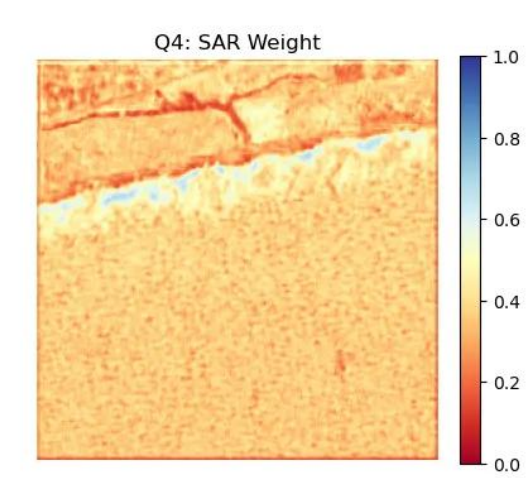
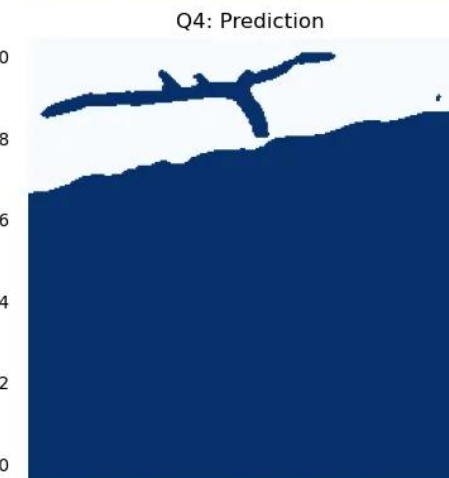
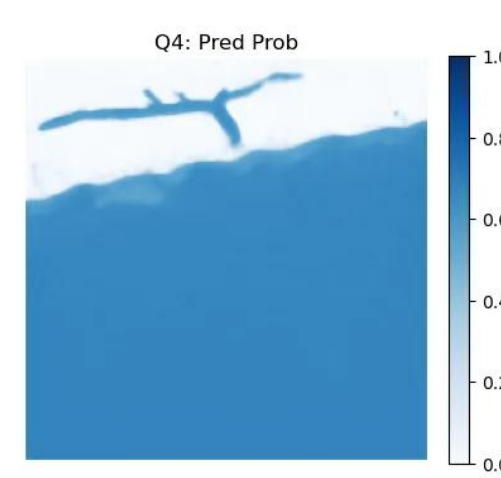
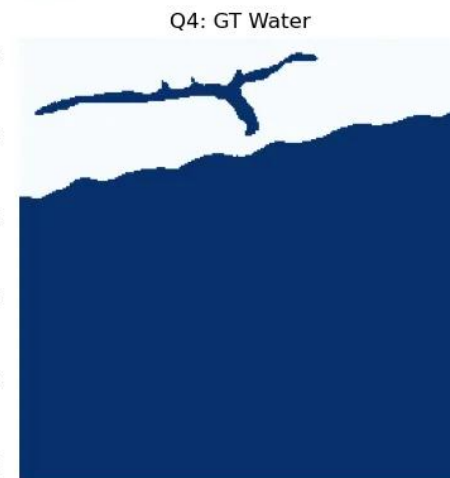
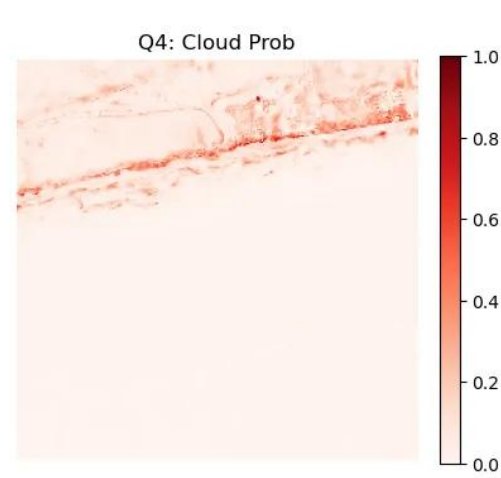
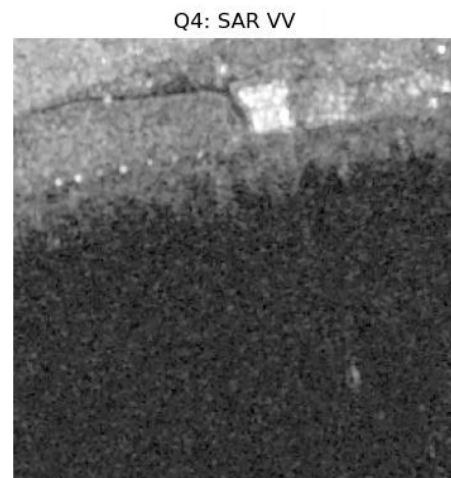
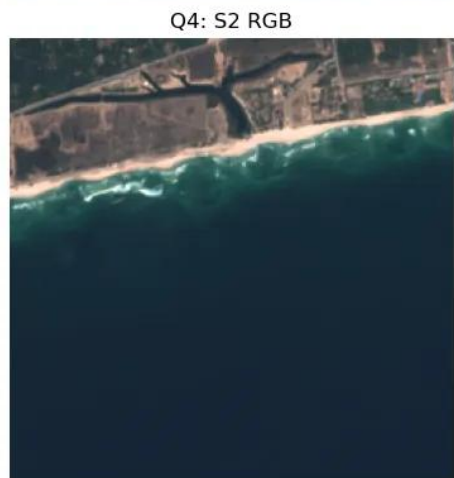
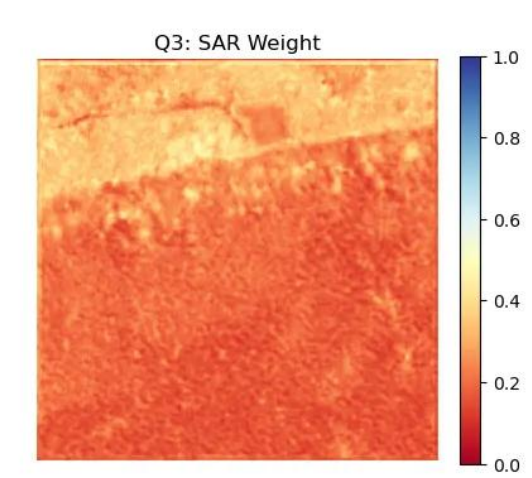
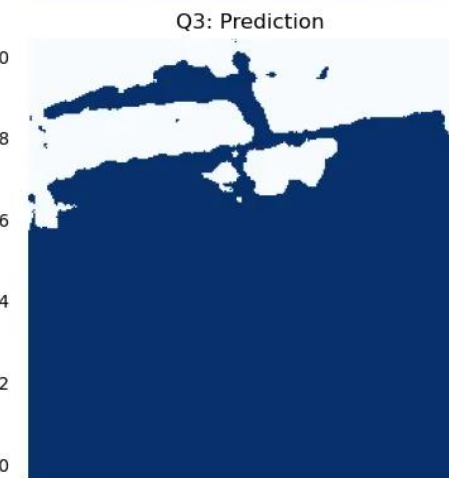
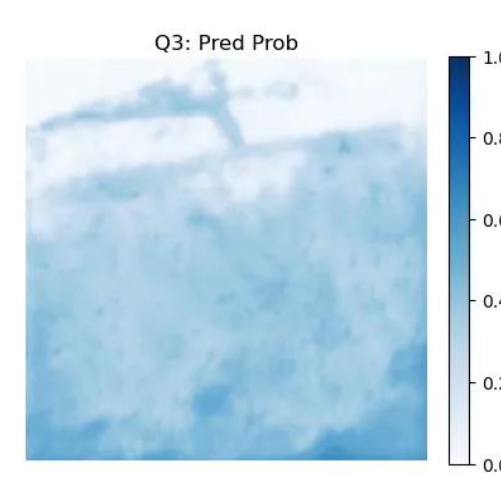
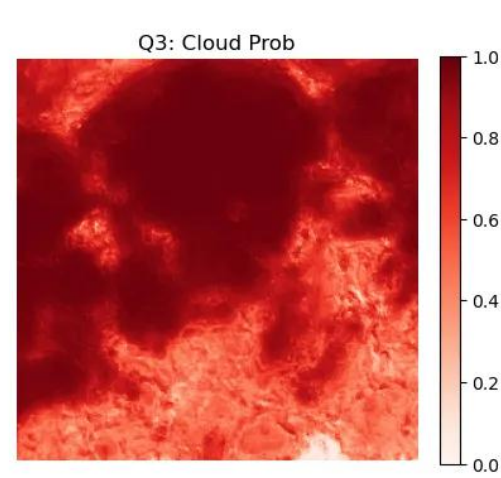
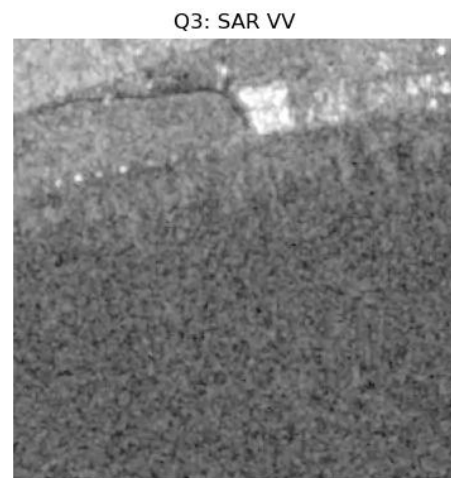
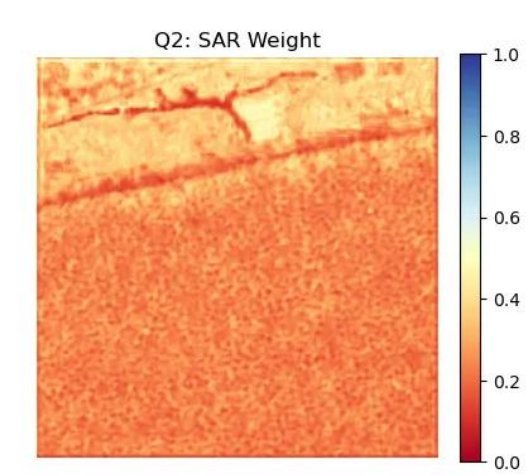
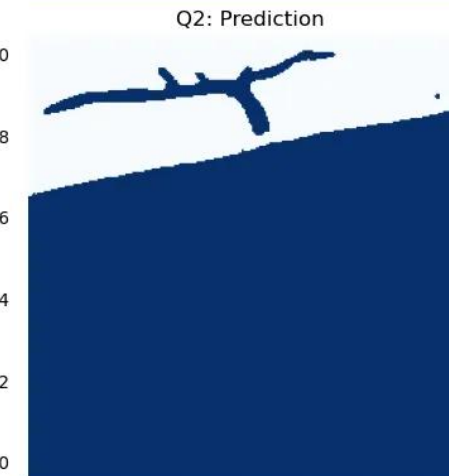
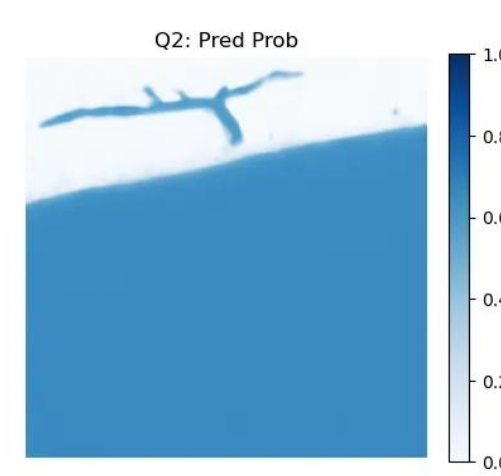
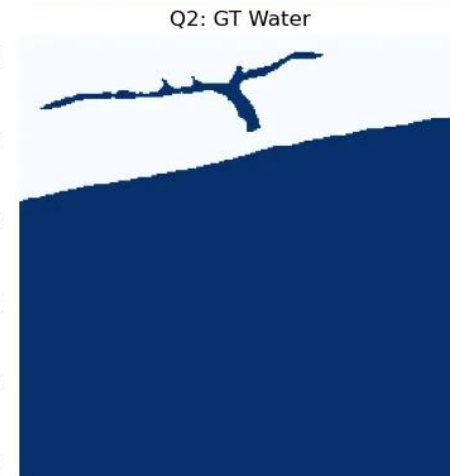
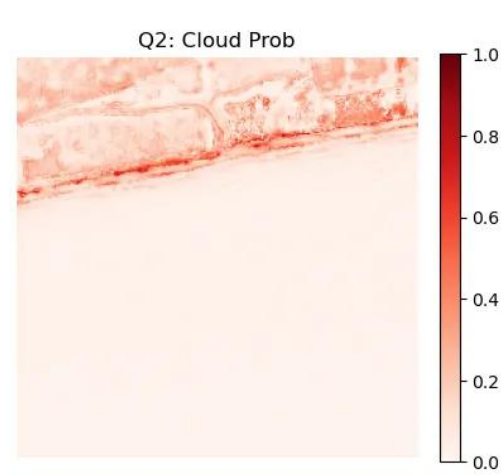
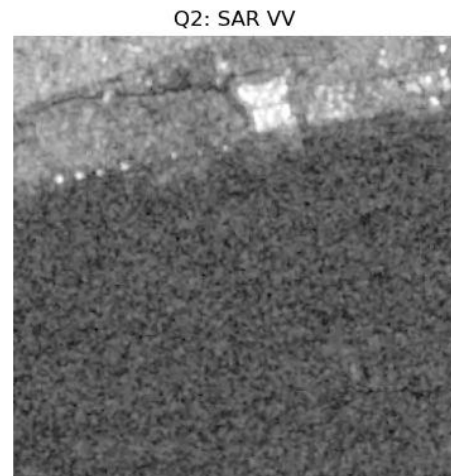
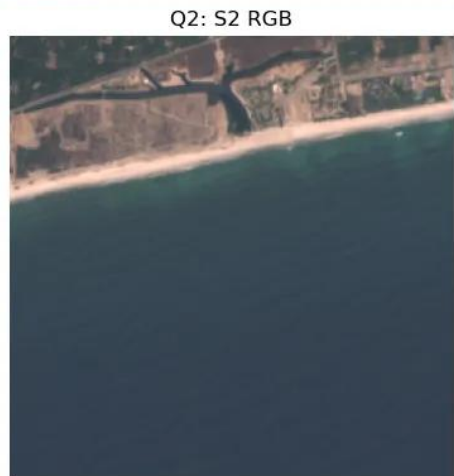
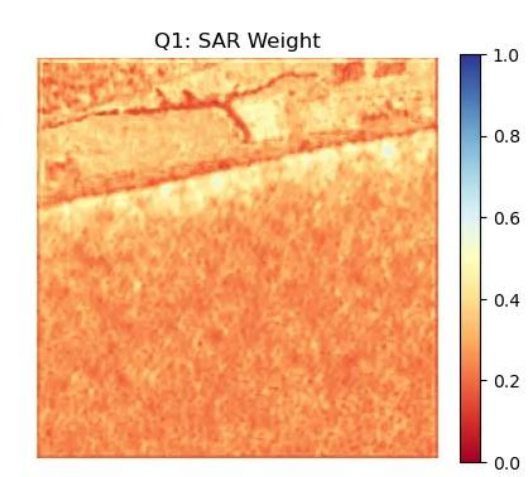
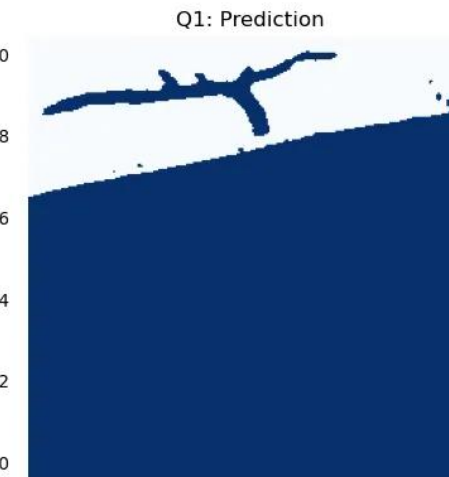
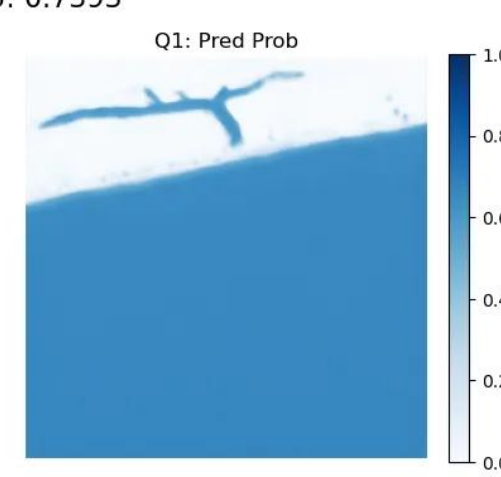
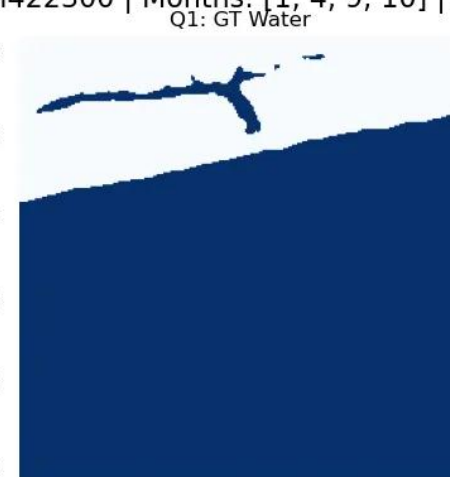
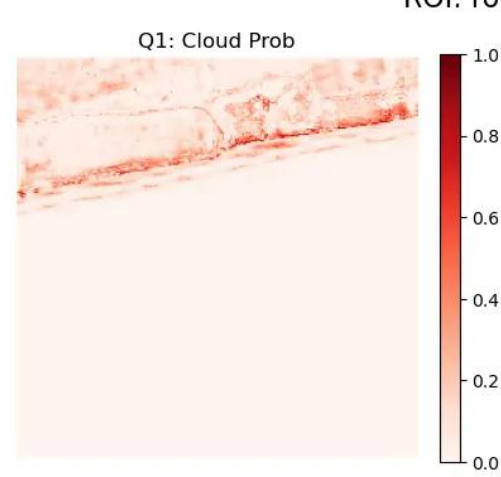
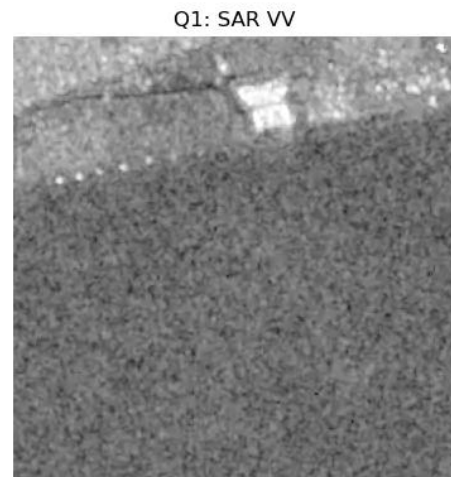
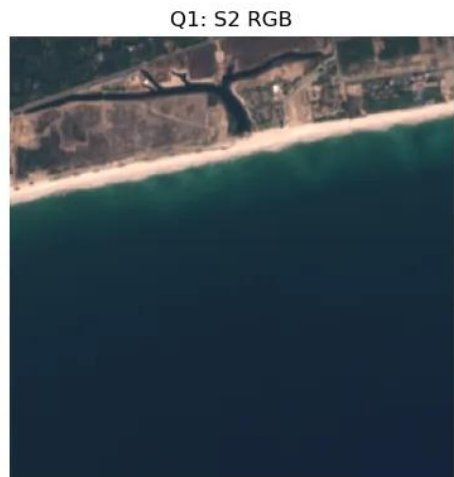








ROI: roi422300 | Months: [1, 4, 9, 10] | IoU: 0.7393

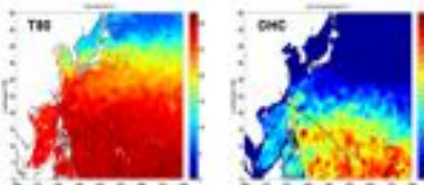


0.Introduction

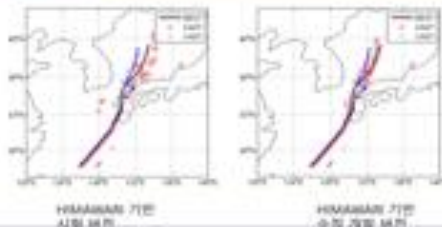
Environmental task

Source : NMSC Information

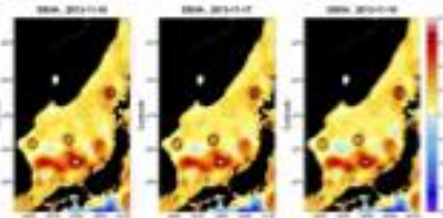
Typhoon/Ocean



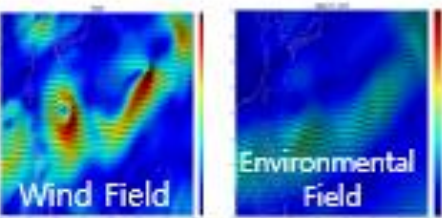
Estimating the intensity change of tropical cyclone



Objective and automatic technique to analyze tropical cyclone (TC) information

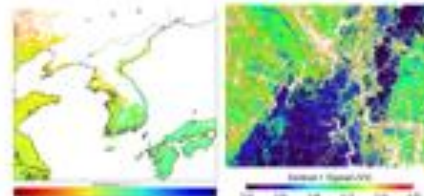


Satellite-based ocean environment monitoring and analysis

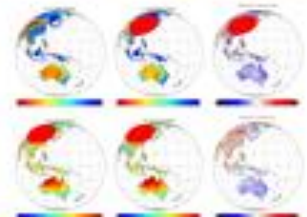


Sea surface wind and 3-D wind field analysis

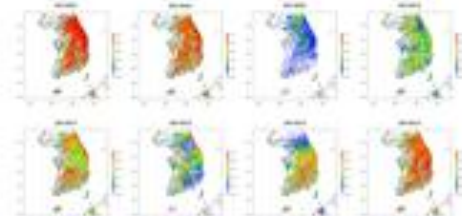
Hydrology/Environment



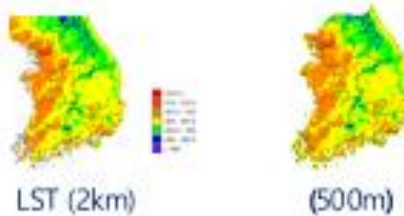
Soil moisture & flood



Drought

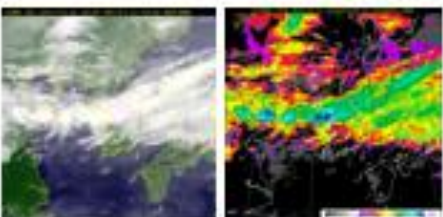


Forest Fire

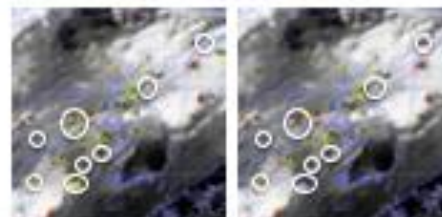


Blending, Downscaling, Validation Strategy

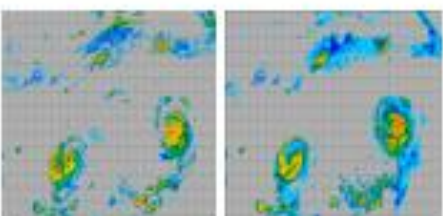
Nowcasting



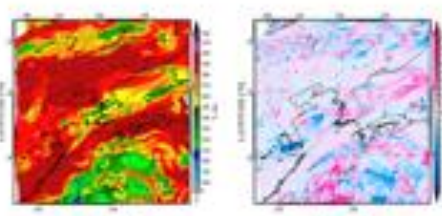
Satellite-based objective cloud analysis



Convective cloud life cycle monitoring and analysis

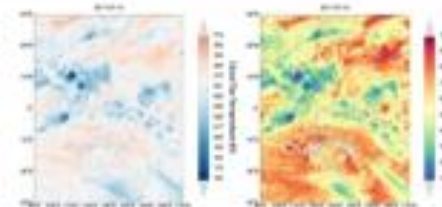


Multi-satellite sensor precipitation merging and analysis

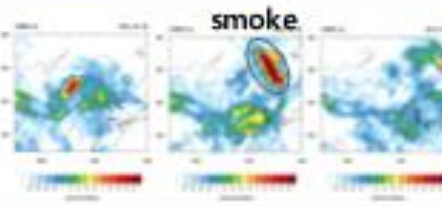


Improvement of meteorological satellite forecasting production technology

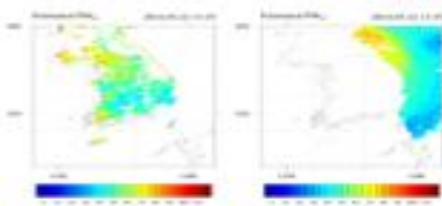
Climate/Environment



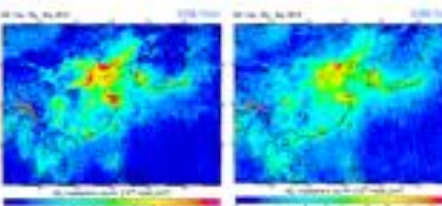
Climate change and environmental meteorological monitoring plan



Environmental Meteorological monitoring and analysis technology development plan



Dust (peak height/concentration)



Research on satellite data utilization technology for environmental meteorological prediction model